SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304
Course Title: AI Assisted Coding
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Lab Objectives

- Apply advanced prompt engineering techniques
- Generate programs using structured prompts
- Improve accuracy of AI-generated code
- Validate and refine outputs

Task 1: Leap Year using Prompt Engineering

Prompt: Write a Python program to check whether a year is a leap year.

Python Code

```
year = int(input("Enter year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not Leap Year")
```

Task 2: Number Conversion

Prompt: Convert temperature from Celsius to Fahrenheit.

Python Code

```
c = float(input("Enter Celsius: "))
f = (c * 9/5) + 32
print("Fahrenheit:", f)
```

Task 3: Vowel Count

Prompt: Write a program to count vowels in a string.

Python Code

```
text = input("Enter string: ")
count = 0

for ch in text.lower():
    if ch in "aeiou":
        count += 1

print("Vowels:", count)
```

Task 4: File Handling

Prompt: Write a Python program to read a file and count words.

Python Code

```
file = open("sample.txt", "r")
data = file.read()

words = data.split()
print("Word count:", len(words))

file.close()
```

Conclusion

Advanced prompt engineering improves code generation quality.

Providing clear instructions and structured prompts leads to better results.